

BAR-SAV for Source Address Validation and BGP Prefix Filtering

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BAR-SAV:

SAV using BGP UPDATE messages, ASPA, and ROA

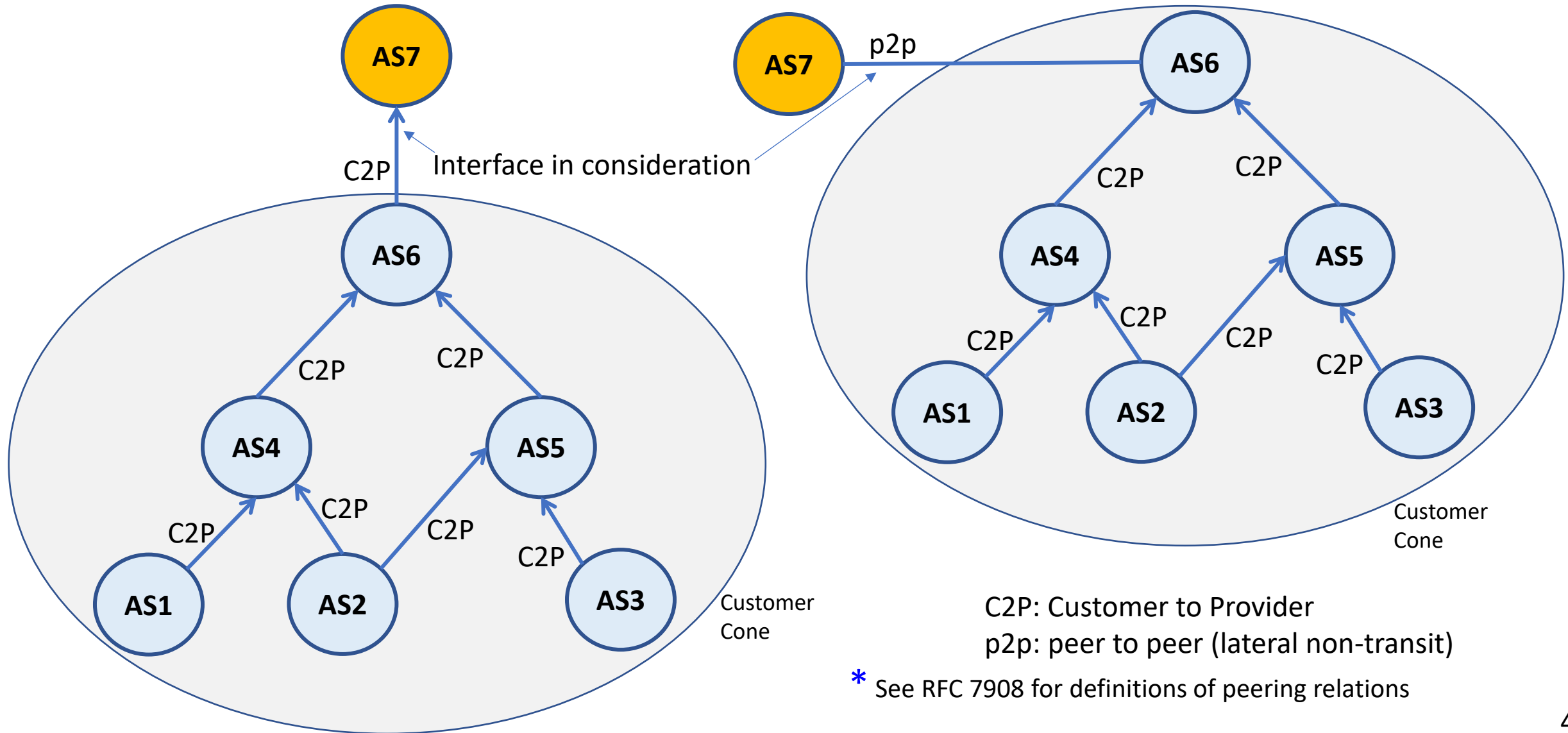
- A method for mitigating IP address spoofing attacks without inadvertently blocking legitimate traffic in complex networking topologies
- Method for ISP to discover its Customer Cone (CC)
 - What ASes are in the CC?
 - What prefixes are originated in the CC?
 - Useful for ISP BGP Prefix Filtering

Motivation and Summary

- Much interest seen in the community to improve Source Address Validation (SAV) techniques (e.g., RFC 8704, IETF SAVNET WG)
- There are attempts to further improve upon EFP-uRPF [RFC 8704]
- Proposed new BAR-SAV method makes complementary use of BGP UPDATES, ASPAs, and ROAs
- BAR-SAV advances the technology for SAV table (allowlist) design
 - ✓ Significantly improves the ability to detect hidden prefixes
 - ✓ Provides a solution to the CDN/Direct Server Return (DSR) problem
- Useful for ISP Customer Cone Discovery and BGP Prefix Filtering
- No changes to protocol on the wire
- Offers immediate benefits to early adopters

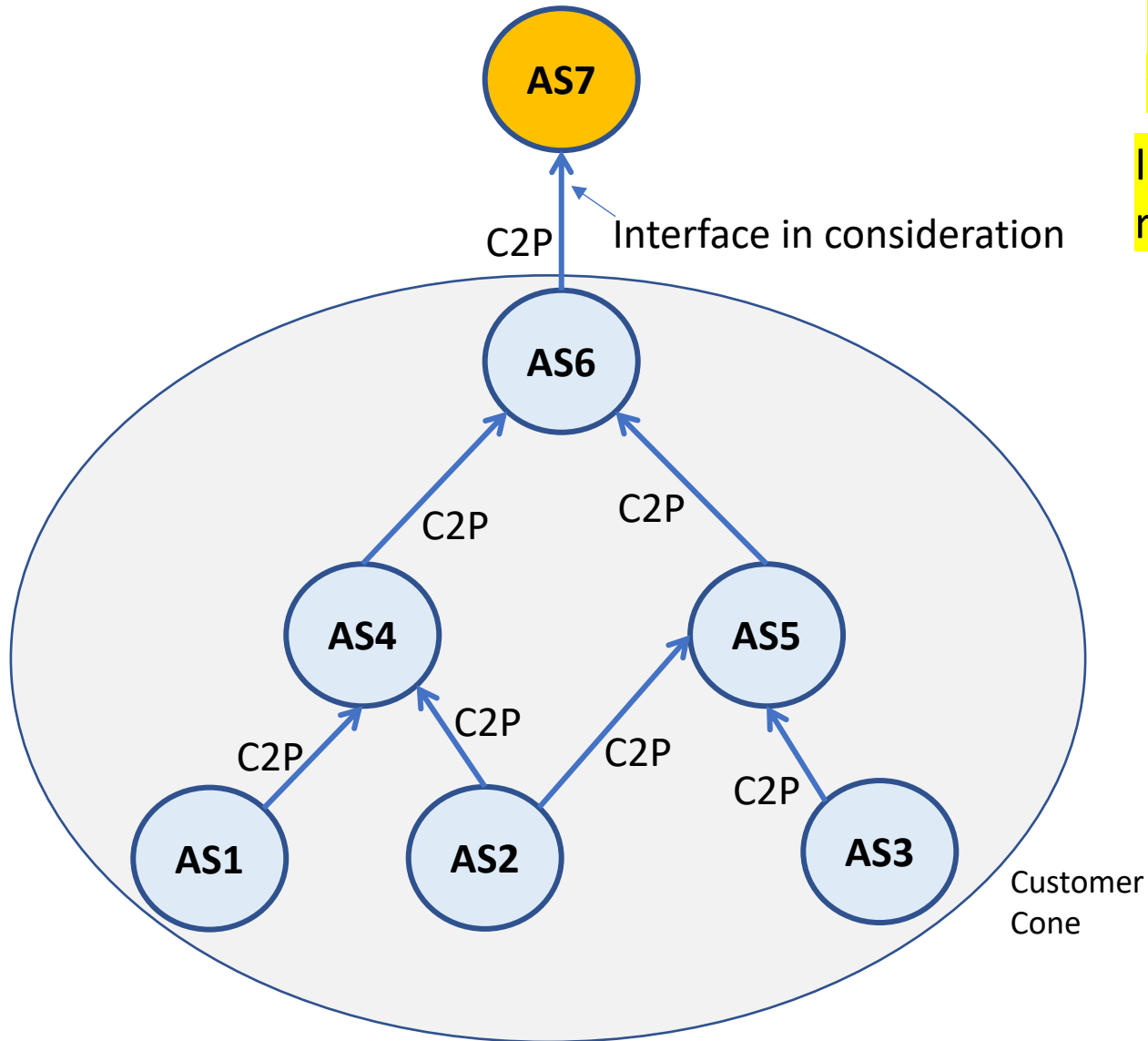
Goal: Construct Permissible Ingress Prefix List for SAV (at AS7)

The methodology is the same for a Customer or Lateral (i.e., non-transit) Peer* Interface



SAV Using Only ASPA and ROA (Procedure X)

Construction of Permissible Ingress Prefix List for SAV (at AS7)



When ASPA and ROA adoption is ubiquitous (in the future)

ISP may use Procedure X on customer interfaces if it requires all its customers to register ROAs and ASPAs

- Obtain the set of ASNs in the Customer's customer cone (CC) using ASPAs
- Gather all prefixes in ROAs associated with the ASNs found in Step A. Keep only the unique prefixes.
- The set computed in Step B is the permissible prefix list* for SAV for the interface in consideration.

*Can be used for ISP prefix filtering as well

But there will be...

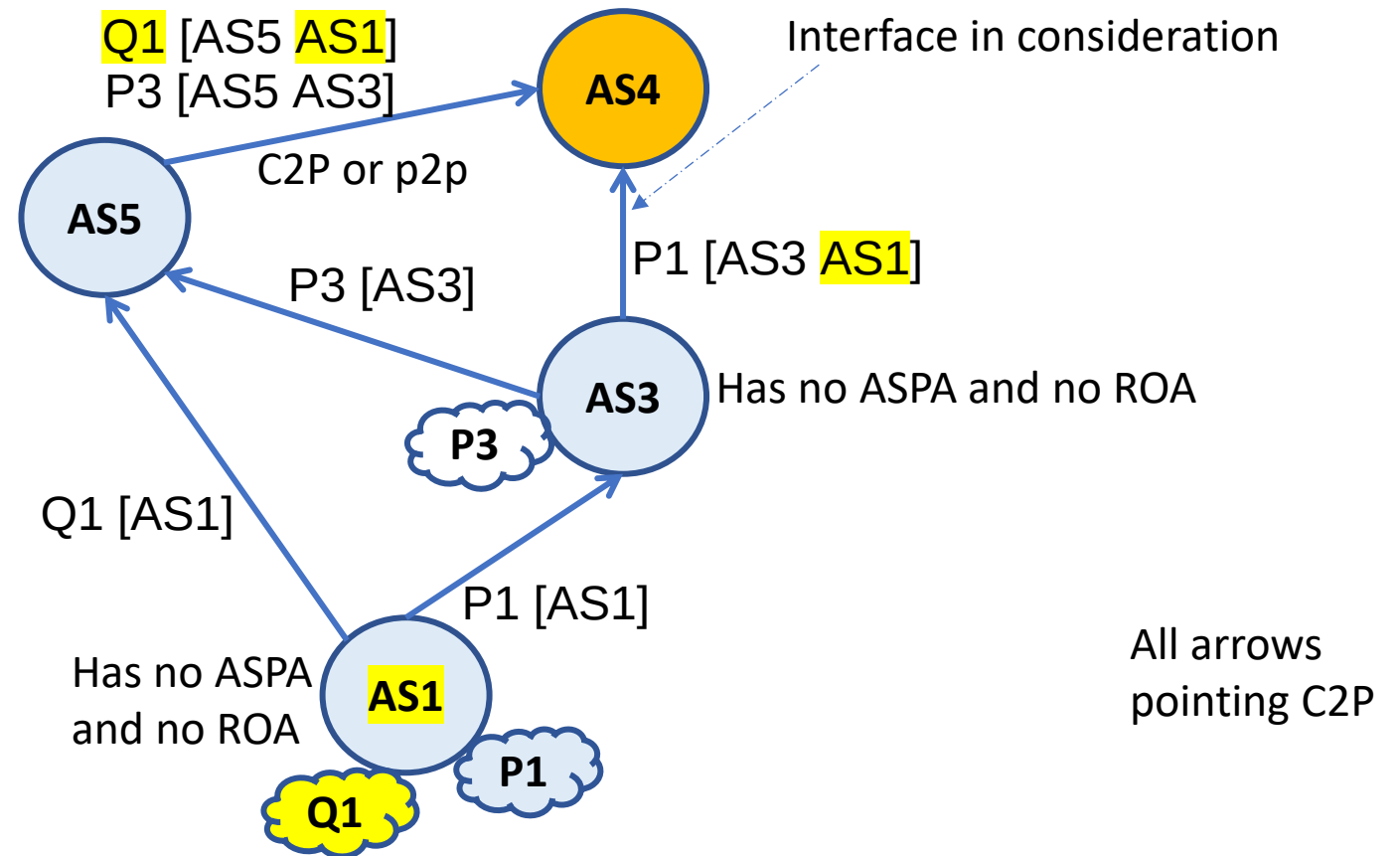
Partial deployment of ROAs and ASPAs for some time

- During that period...
 - ✓ BAR-SAV compensates
 - ✓ Makes complementary use of BGP UPDATES, ASPA, and ROA
 - Incorporates a refined version of EFP-uRPF*

* Enhanced Feasible Path uRPF (EFP-uRPF) [RFC 8704]

Enhanced Feasible Path uRPF (EFP-uRPF) [RFC 8704] in a Nutshell

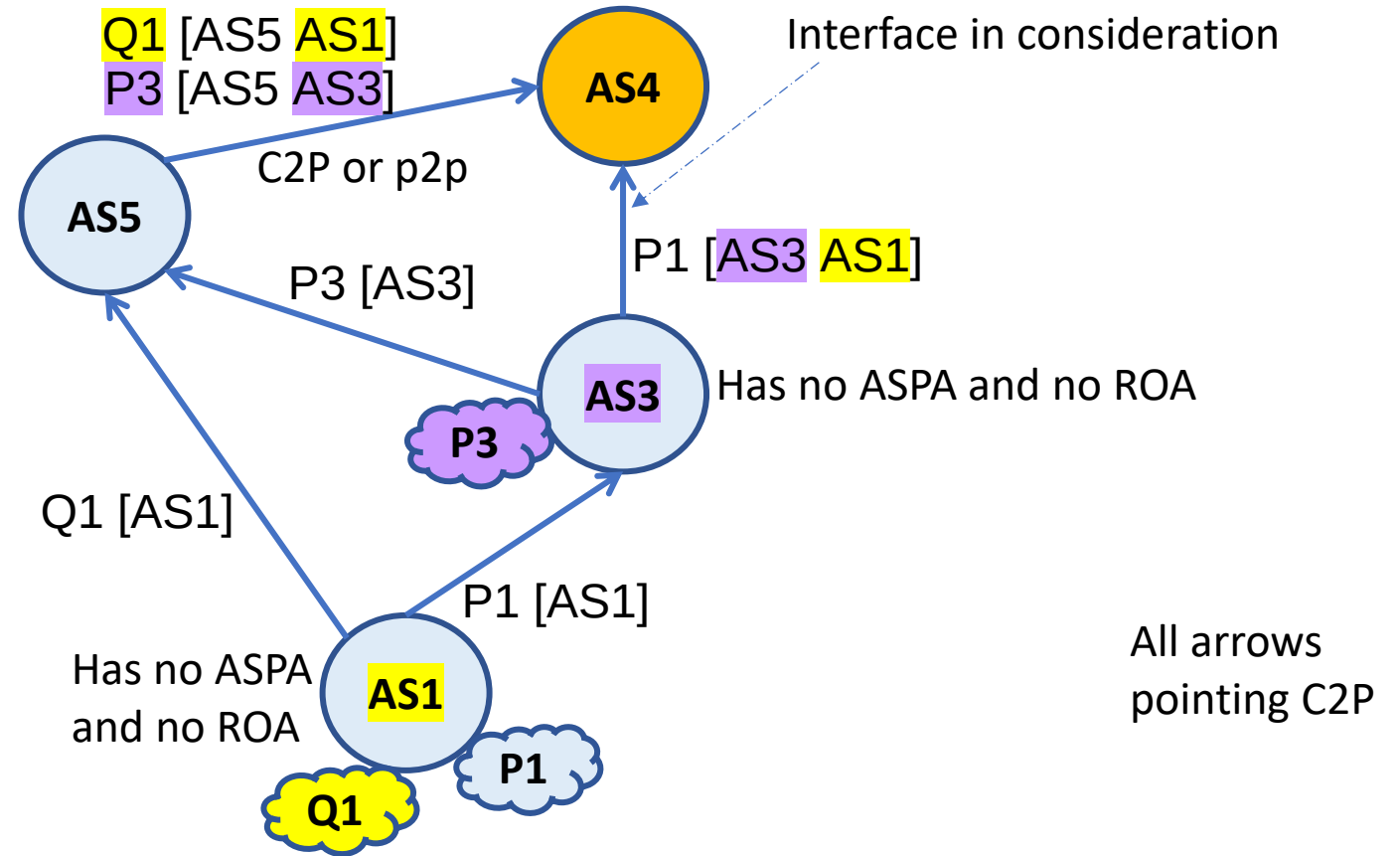
- Careful consideration of asymmetric routing and packet forwarding
- Prefixes P1 and Q1 are captured by Alg. A of RFC 8704
- P3 is not captured
- EFP-uRPF considered only the origin AS in AS_PATHs for correlations
- Did not consider the ASes in the middle to capture more hidden prefixes



EFP-uRPF = Enhanced Feasible Path uRPF

Refined Version of EFP-uRPF [RFC 8704] Incorporated into BAR-SAV

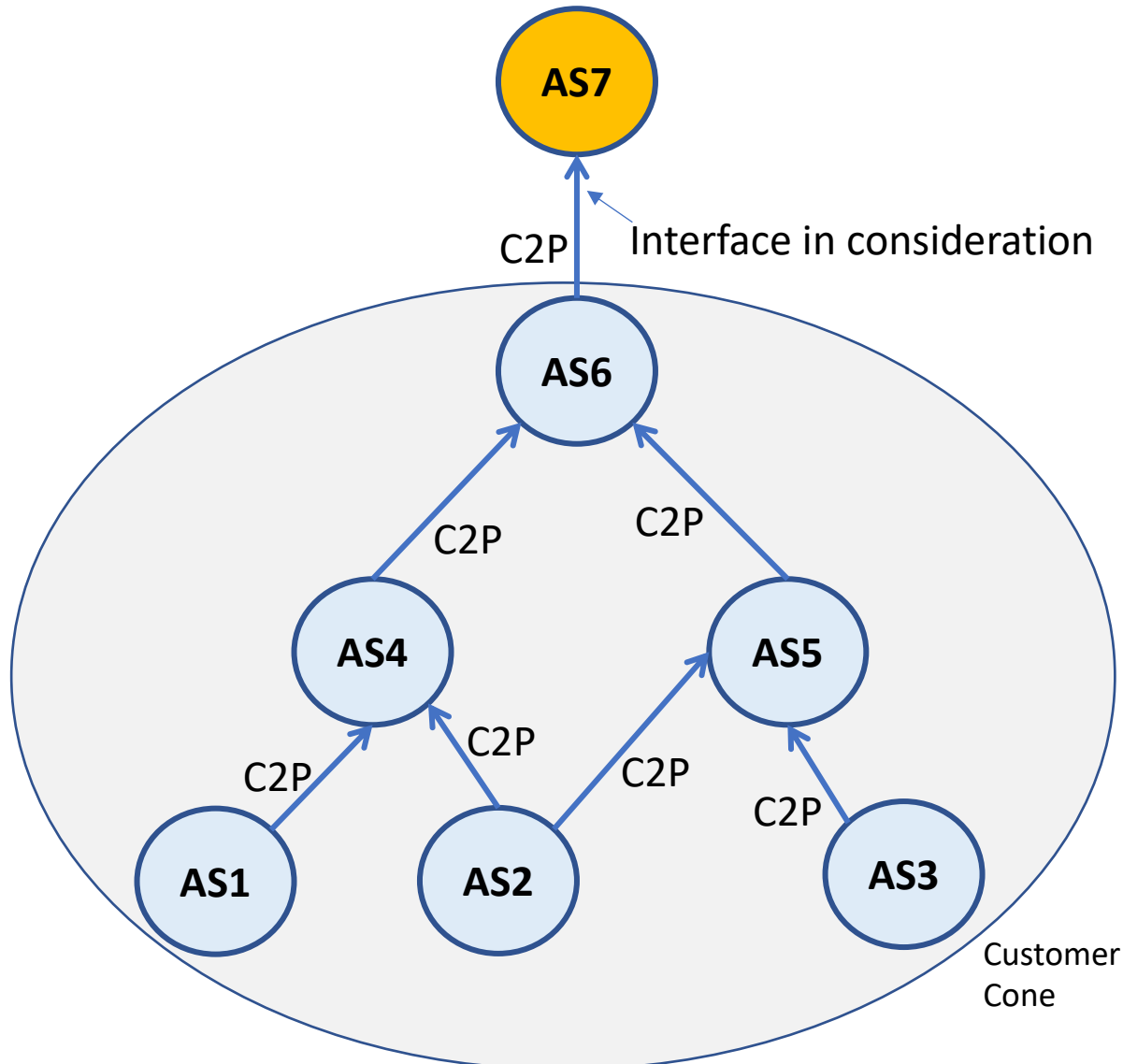
- BAR-SAV makes more efficient use of AS_PATHs for discovering more ASes and prefixes in the CC
- Considers ASes in the middle of the path (e.g., AS3) as well for correlations, not just the origin ASes
- Unlike EFP-uRPF, BAR-SAV captures P3 also



- Much better detection of “Hidden” prefixes in multihoming scenarios by BAR-SAV

SAV using BGP UPDATEs, ASPA, and ROA (BAR-SAV)

Construction of Permissible Ingress Prefix List for SAV (at AS7)

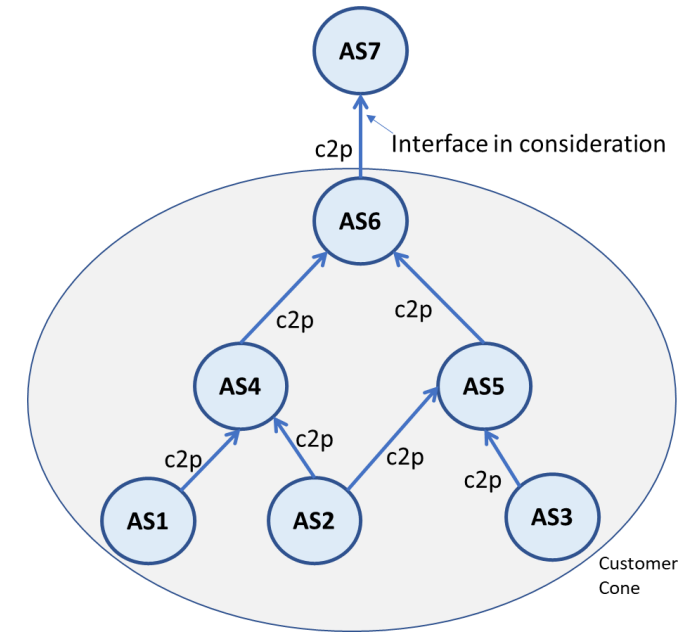


Applicable in the period when ASPA and ROA adoption is not ubiquitous

- Obtain the set of ASNs in the Customer's customer cone (CC) using ASPAs and AS_PATHs
- Gather all prefixes in ROAs associated with the ASNs found in Step A.
- Gather all prefixes in BGP UPDATE messages with origins among ASNs found in Step A.
- Combine sets found in Steps B and C. Keep only the unique prefixes. This is the permissible prefix list for SAV for the interface in consideration.

A Note on Customer Cone Computation

- One should *not* compute a customer cone by separately processing ASPA data and AS_PATH data and then merging the two sets of ASes at the end. Doing so is likely to miss ASes from the customer cone.



- Instead, both ASPAs and AS_PATHs should be used to iteratively expand the discovered customer cone. When new ASes are discovered, both ASPA and AS_PATH data should be used to discover customers of those ASes. This process is repeated for newly discovered customer ASes until there are no new ASes to be found.

The next 3 slides illustrate the details
of how BAR-SAV works

How BAR-SAV Works

Finding All ASes and Prefixes in Customer's (or Peer's) Customer Cone
Using BGP Announcements (as seen at AS4), ASPA, and ROA

ASPA:

AS3 {AS4,
AS9}

AS5 {AS1}

AS6 {AS1}

AS8 {AS2}

ROAs:

P2a AS2

P5 AS5

P6 AS6

P8 AS8

*Customer
Cone of AS3*

Interface in
consideration

P6 [AS3 AS1 AS6]
P7 [AS3 AS1 AS7]

AS4

P2C or p2p or C2P

P2 [AS9 AS3 AS2]
...

AS9

Has ROA

Pn

No ROA

Pn

At AS4's interface with AS3:

Visible in ASPA
and AS_PATH

ASn

Visible only
in AS_PATH

ASn

Visible only
in ASPA

ASn

Not visible in ASPA
(indirectly visible in
AS_PATH; BAR-SAV
helps find it)

ASn

C2P: Customer to Provider
P2C: Provider to Customer
p2p: peer to peer (lateral non-transit)

Finding All ASes in the CC of AS3 using BGP AS_PATH and ASPA

INPUTS

ASPA:

AS3 {AS4, AS9}

AS5 {AS1}

AS6 {AS1}

AS8 {AS2}

ROAs:

P2a AS2

P5 AS5

P6 AS6

P8 AS8

BGP UPDATE AS_PATHs:

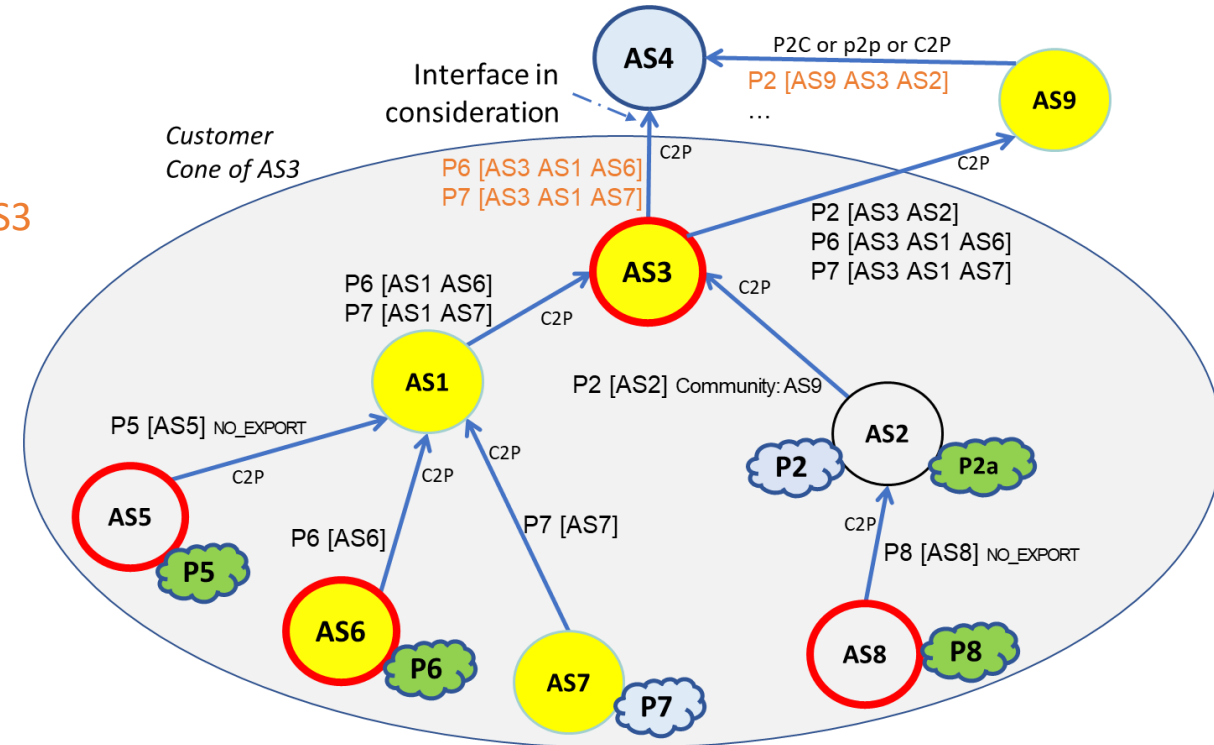
Interface in Consideration: **AS3**

P6 [**AS3** AS1 AS6]

P7 [**AS3** AS1 AS7]

Other Interfaces:

P2 [AS9 AS3 AS2]



OUTPUT

Iteration	Customer Cone	New ASes from ASPA	New ASes from AS_PATH
1	AS3	None	P6 [AS3 <u>AS1</u> AS6] → AS1 P7 [AS3 <u>AS1</u> AS7] → AS1 P2 [AS9 AS3 <u>AS2</u>] → AS2
2	AS3, AS1 , AS2	AS5 { AS1 } → AS5 AS6 { AS1 } → AS6 AS8 { AS2 } → AS8	P6 [AS3 AS1 <u>AS6</u>] → AS6 P7 [AS3 AS1 <u>AS7</u>] → AS7
3	AS3, AS1, AS2, AS5 , AS6 , AS8 , AS7	None	None

Finding All Prefixes in the CC of AS3 using BGP Routes and ROA

INPUTS

ASPsAs:

AS3 {AS4}
AS3 {AS9}
AS5 {AS1}
AS6 {AS1}
AS8 {AS2}

ROAs:

P2a AS2
P5 AS5
P6 AS6
P8 AS8

BGP UPDATE AS_PATHs:

Interface in Consideration: AS3

P6 [AS3 AS1 AS6]

P7 [AS3 AS1 AS7]

Other Interfaces:

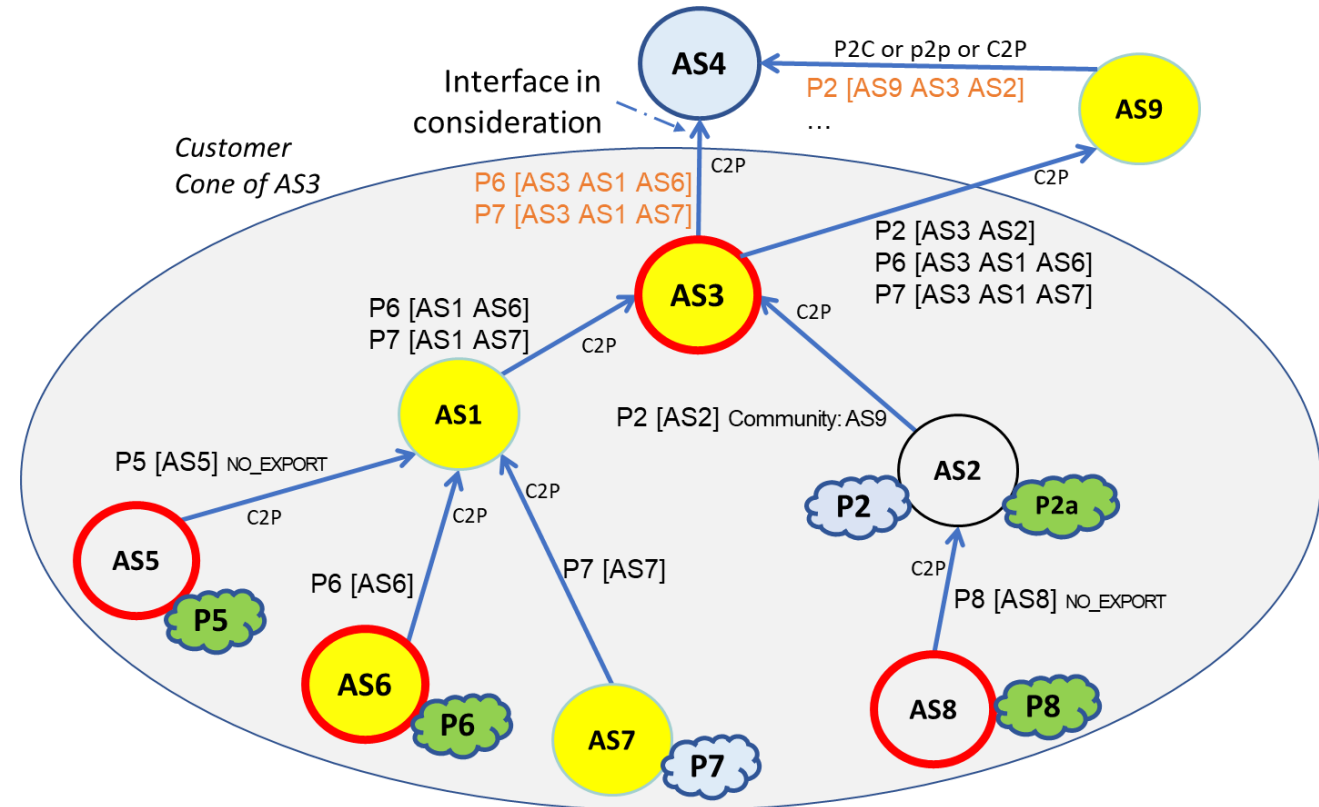
P2 [AS9 AS3 AS2]

Customer Cone

AS1, AS2, AS3, AS5, AS6, AS7, AS8

OUTPUT

ASN	Prefixes from ROA	Prefixes from BGP
AS1		
AS2	(<u>P2a</u> AS2) → P2a	<u>P2</u> [AS9 AS3 AS2] → P2
AS3		
AS5	(<u>P5</u> AS5) → P5	
AS6	(<u>P6</u> AS6) → P6	<u>P6</u> [AS3 AS1 AS6] → P6
AS7		<u>P7</u> [AS3 AS1 AS7] → P7
AS8	(<u>P8</u> AS8) → P8	



SAV Prefixes

P2, P2a, P5, P6, P7, P8

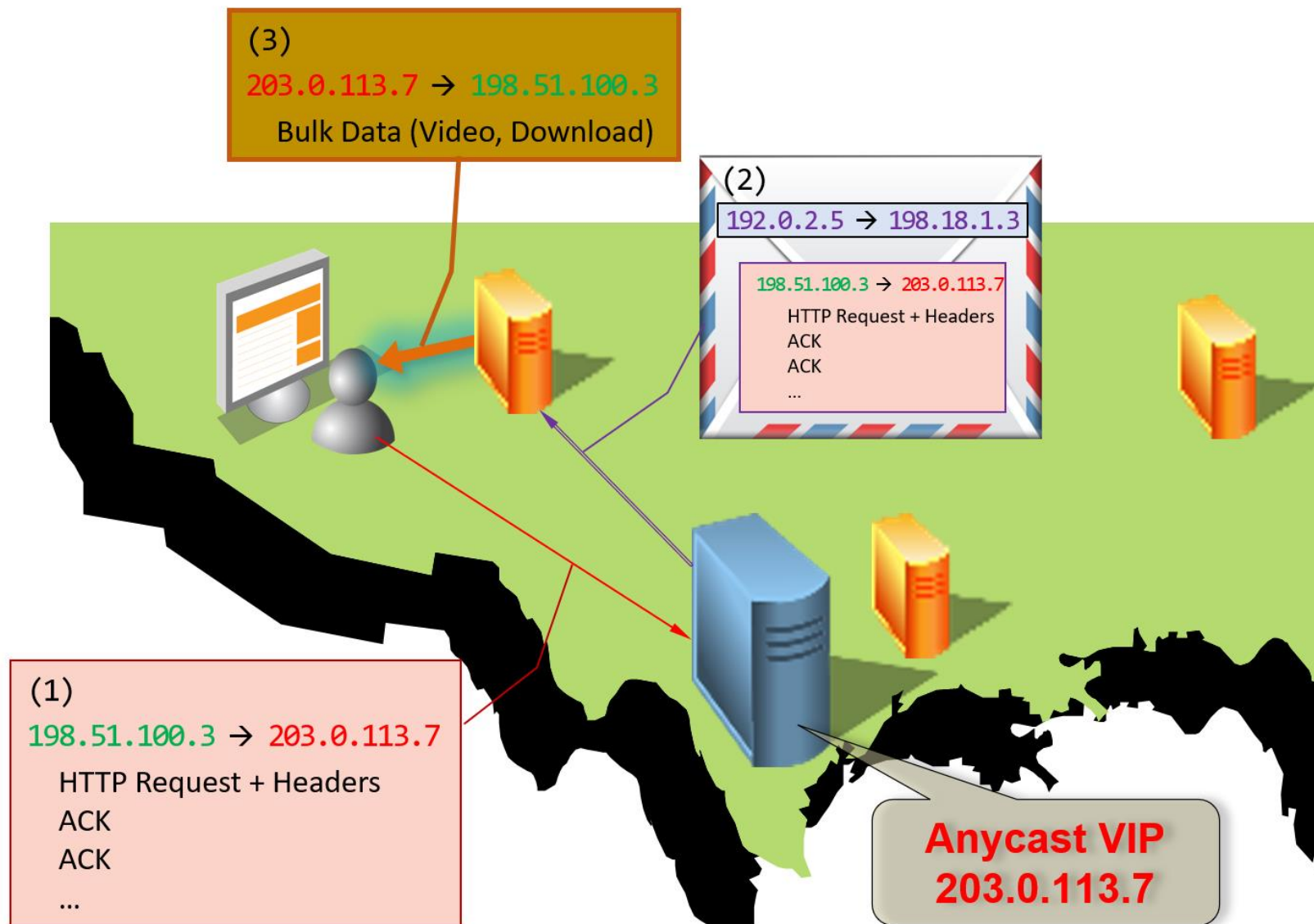
Detailed Description of the BAR-SAV Procedure

- See Sections 3 and 4 in the BAR-SAV draft

<https://datatracker.ietf.org/doc/draft-ietf-sidrops-bar-sav/>

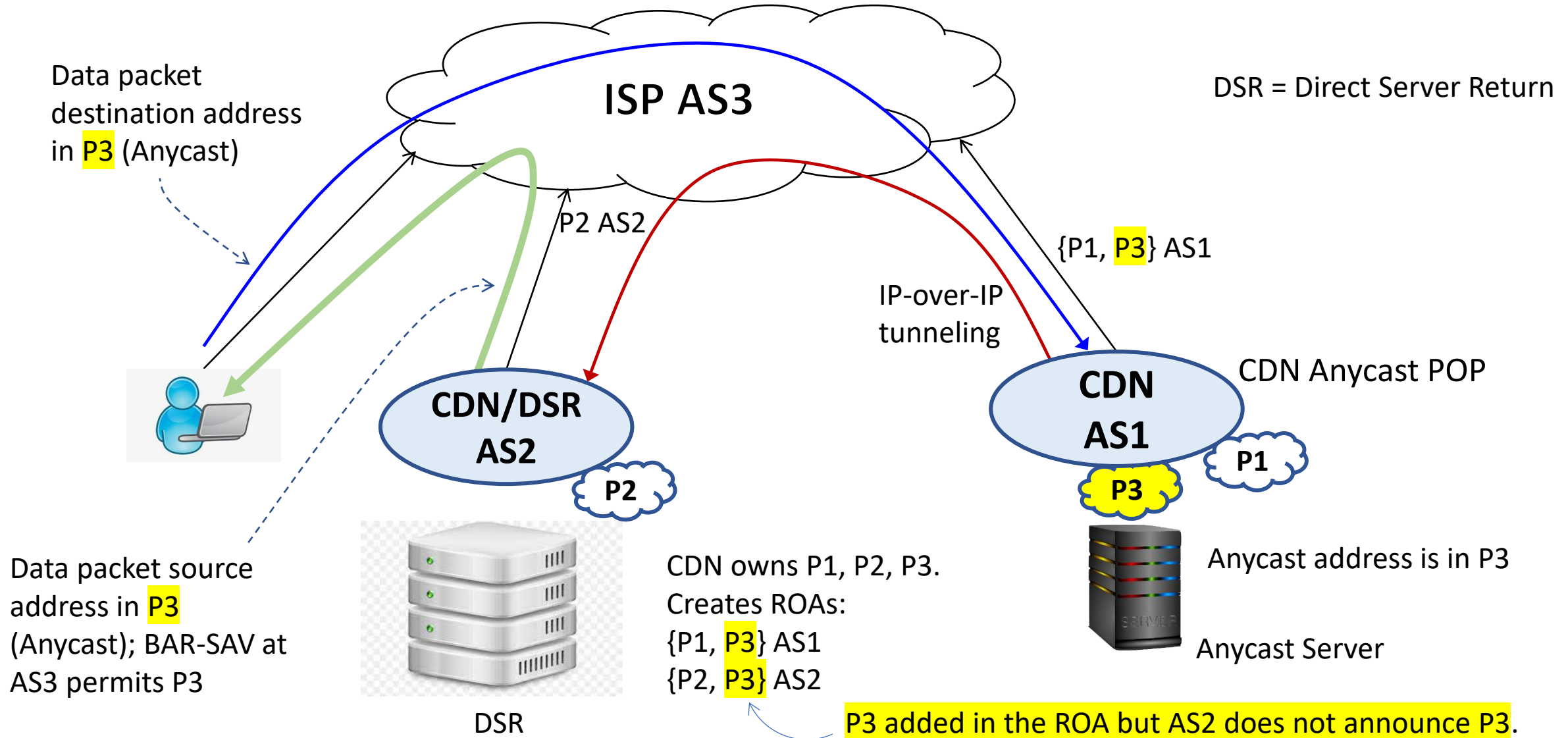
Anycast/Edge Hybrid – Direct Server Return

1. Anycast POPs lookup “best” edge POP for each new connection (using the actual user IP)
2. Anycast POPs tunnel packets to edge POPs
3. Edge servers send data to users directly – Direct Server Return (DSR)



Content Delivery Network (CDN) Application

Example of how the BAR-SAV method solves the DSR blocking problem



NO_EXPORT and DSR Scenarios

- BAR-SAV utilizes ASPA to uncover hidden NO_EXPORT prefixes
- It utilizes ROA to uncover hidden DSR (Direct Server Return) prefixes
- Strongly recommend network operators to
 - Register ASPA in case of origination of routes with NO_EXPORT
 - And/or advertise a covering less-specific prefix also without NO_EXPORT to the same providers
 - Register ROA in case the unannounced DSR anycast prefix scenario applies to their AS

Best Practices Considering RPKI Propagation Delay

- Existing prefix with ROA is split and more specific prefixes under it are announced
 - New ROA is propagated
 - BCP recommendation to operator is to use make-before-break principle
 - E.g., withdraw the less-specific prefix and old ROA only after the new ROA has propagated
- AS has a new transit provider and ASPA is updated
 - Again, use the make-before-break principle
 - E.g., continue to announce routes via old transit provider until new ASPA has propagated

Traffic Origin Authorization (TOA)

- New RPKI object proposed in the IETF
- TOA to be used when a prefix owner intends to not announce a prefix but use it to source traffic from a specified AS
 - Semantic details and implications being discussed in the SAVNET WG
 - Currently BAR-SAV recommends registering ROA even in these cases
- BAR-SAV can incorporate TOA if it is advanced in the IETF

ISP BGP Prefix Filtering and Know Your Customer Cone

- BAR-SAV finds all originated prefixes and ASes in the CC
- ISP can implement prefix filtering using the discovered list of prefixes
 - On interfaces facing customers
 - On interfaces facing lateral peers
- Even if not doing SAV or prefix filtering, ISP can still use BAR-SAV to have knowledge of
 - the ASes in the CC
 - the list of prefixes originated in the CC

• **Thank you!**

Questions?

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